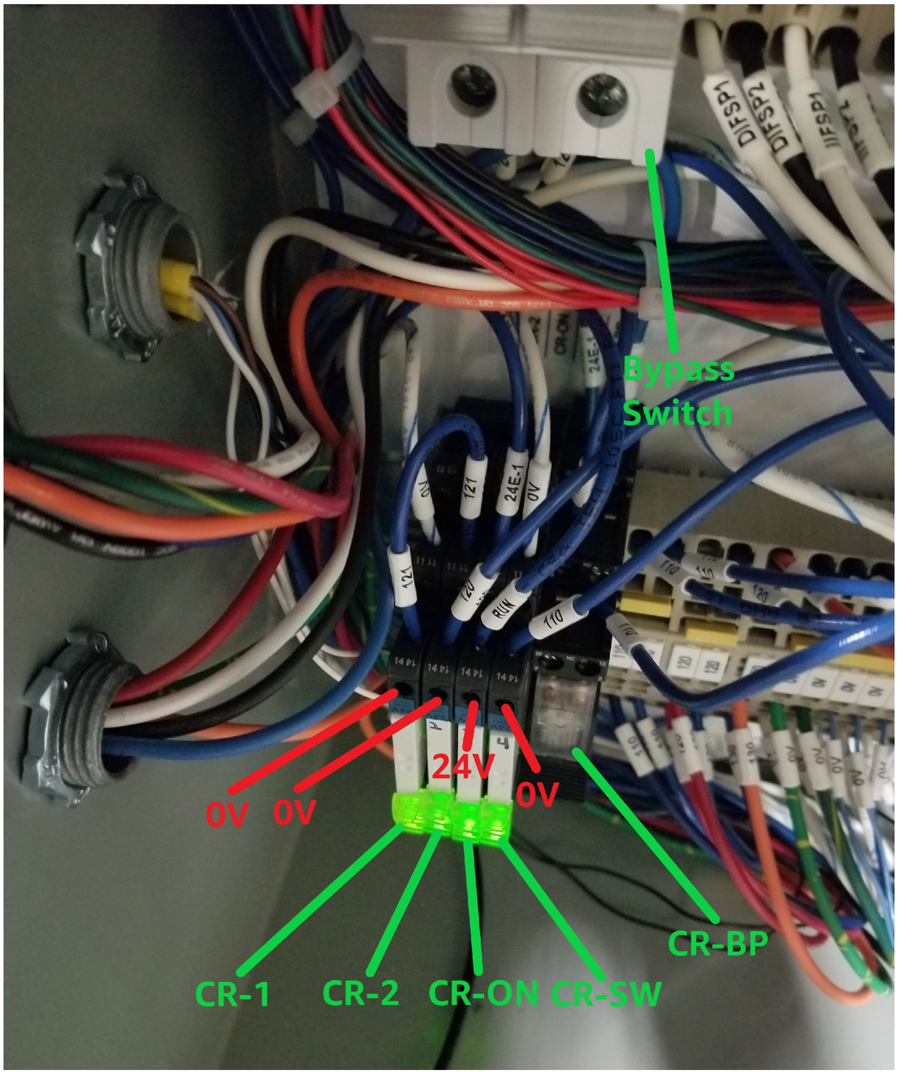
Troubleshooting MOD-LINX Conveyors



View of relays in Master cabinet when the line is running.

Contents:

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Getting Familiar with the System

There are 2 schematics we need to look at.

One is PS20115MLM4-2.pdf and the other is PS10115MLC2-2.pdf.

Press and hold the **Control** button while double clicking the links above to view those files.

We will be looking at these while we discuss how the system works.

Note: *Voltage measurements in the photos are made with respect to common. So the black lead of the meter is on the common bus and the red lead of the meter is on the terminal being measured.*

In the above photo we see the Master (the first and most upstream unit) in the running state.

The relay to the left (CR-1) is lit (energized) when one of the start/stop buttons on this unit is green.

The relay second from left (CR-2) is lit (energized) when the other start/stop button is green.

These two relays are in series so they perform an AND operation along with the rest of the stop/start circuit by providing a path for Wire 120 to get to the Common terminal bus (0 volts DC).

When the relays CR-1 and CR-2 are lit (energized) we should see 0 volts at their terminals marked 14 where Wires 121 and 120 are connected. This is because they connect in series to a common wire at terminal 11 on relay CR-1 when the relays energize.

So jumping Wire 120 in the Master machine to the common bus will signal "both are buttons green" to the rest of the system regardless of the actual state of the buttons.

The third relay from the left (CR-ON) is lit (energized) as a command to run the MDRs on this Master.

This relay passes 24 volts to the RUN Wire when energized.

The RUN Wire provides the 24 volt signal to the run pin on all drives of this machine.

So jumping the RUN Wire to 24 volts will cause the MDRs on this master to run.

When the relay (CR-ON) is energized we should see 24 volts on its terminal 14 because its terminal 11 is connected directly to the 24 volt bus. So if we do not see 24 volts on terminal 14 when the relay is lit then most likely the relay is bad.

The fourth relay from the left (CR-SW) is lit (energized) as a command to run all the other conveyors and this one too. Relay CR-SW becomes energized when every button on every machine in the line is green.

Relay CR-SW connects Wire 110 to the common bus inside the Master machine. Wire 110 is connected to the common side (terminal A2) of every CR-ON relay of every machine in the line. So this is a very long wire that goes through every machine via the yellow cables which connect each upstream machine to its downstream neighbor.

So jumping Wire 110 to 0 volts will cause all the machines to run regardless if any of the buttons are red. Only do this in the test environment – never in production because the buttons will not be able to stop the machines.

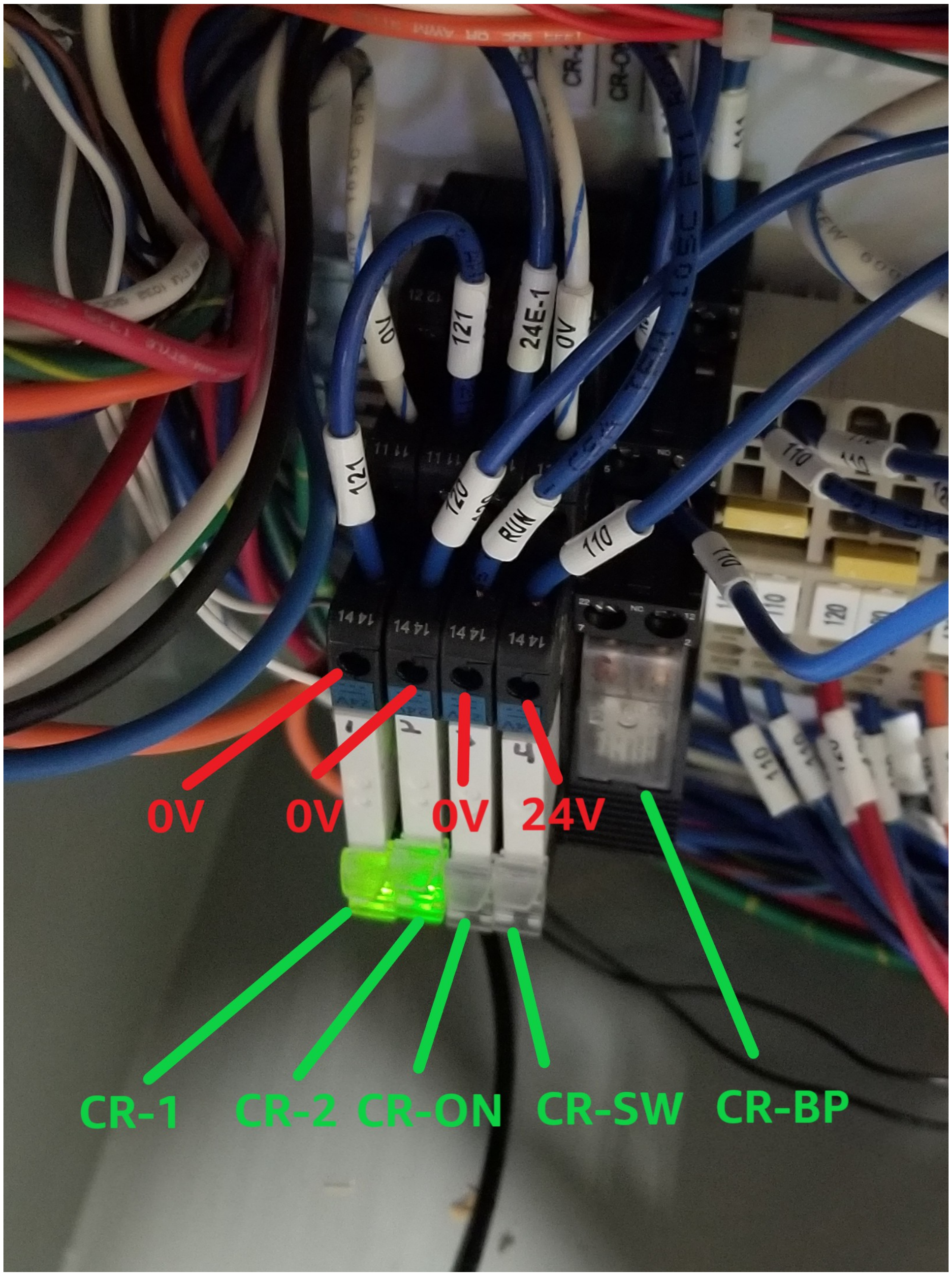
When relay CR-SW in the master cabinet is lit we should always see 0 volts on its terminal 14. If not the relay is probably bad or a wire is loose going to the common bus from terminal 11.

There is a fifth relay all the way to the right (CR-BP). This relay will not energize under normal running or stop conditions unless this is the last conveyor on the line. Since we are looking at the Master and since the master is the first unit of the line, this relay will not energize under normal conditions.

Flipping the bypass circuit breaker/switch will cause this relay to energize if both buttons on the master are green. In that case this machine and this machine only will run.

The purpose of this relay (used only on the last machine in the line) is to send signal back down through all the conveyors to the master so as to signal that all machines have buttons green and are commanded to run.

In other words this relay will only energize for the last unit of the line when all units are signaling “Lights green – good to go”



View of Master relays when an upstream machine has button red and line is stopped.

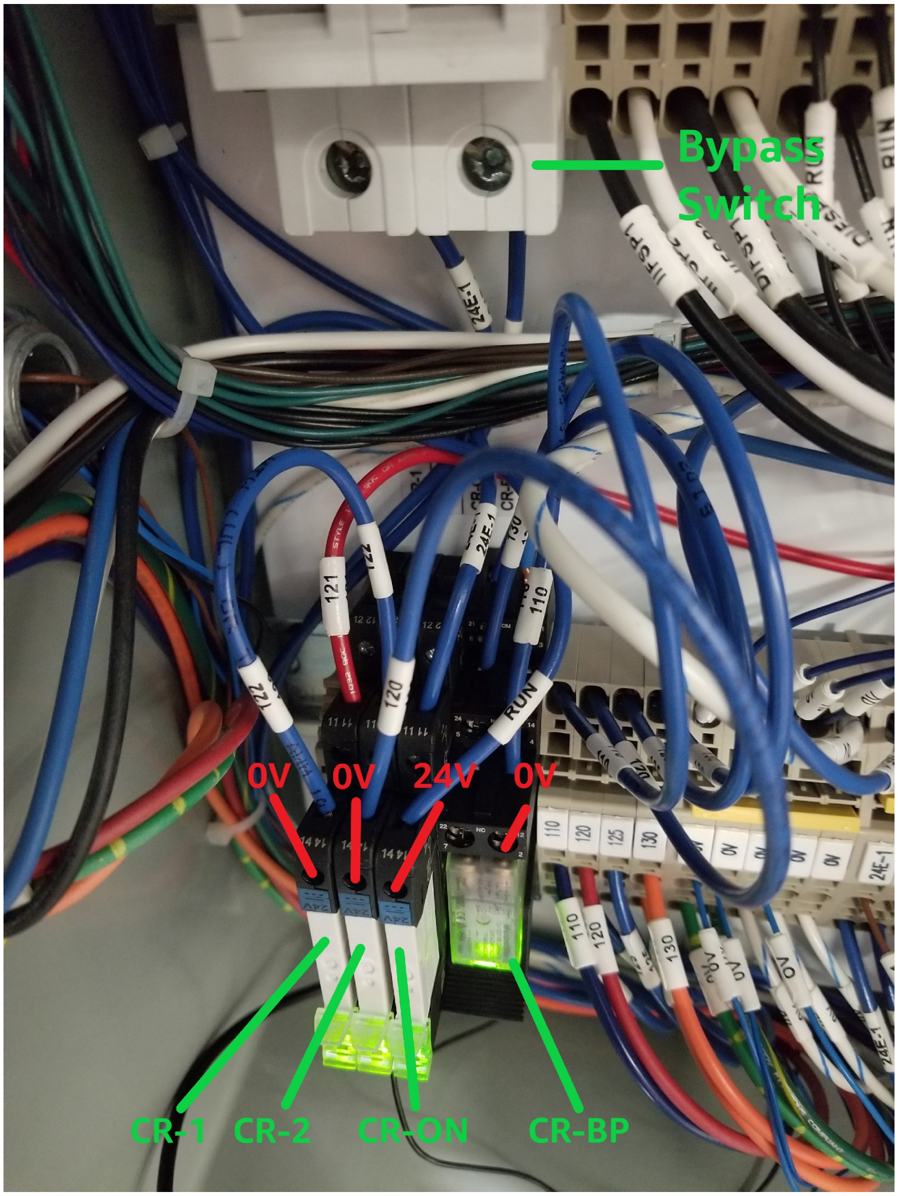
Above we are seeing relays in the Master machine which has buttons green but a downstream child has a button red. In this condition all the machines are stopped.

We see that the first two relays on the left (CR-1 and CR-2) are energized. This means the buttons on this Master are green. Terminals 14 on these relays are at 0 volts so the relays are working correctly because the contact connecting terminals 14 to 0 volts at terminals 11 are supposed to be closed when the relay is energized.

We see that the third relay from the left (CR-ON) is not energized. This means that this machine is not commanded to run. This makes sense because a downstream child has a button in the Red condition. So nothing should run. We have 0 volts on terminal 14 of this relay because the contact is open and cannot receive 24 volts from terminal 11. So there is no 24 volt run signal going to the drives on this machine. That is correct.

We see the fourth relay from the left (CR-SW). This is the all run relay which is connected to Wire 110. It is not energized. This makes sense because nothing should run when a button on any machine is red. Terminal 14 of this relay is at 24 volts. This is correct because when the relay is not energized the contact between terminals 14 and 11 is open so there is no connection to common. The other end of Wire 110 is connected to terminal A2 of every CR-ON relay of every machine in parallel so it gets pulled to 24 volts when there is no common 0v connection. Again, jumping Wire 110 to common will make all the machines run because that energizes every CR-ON relay in every machine. This should only be done for testing – never in production.

We see a fifth relay from the left. This is not energized because the master is not the last unit in the line and also because one of the downstream machines is in the stopped condition.



View of relays in the cabinet of the very last conveyor unit while the line is running.

Above we are looking at the very last machine of the line while the line is running.

So all buttons on all machines are green.

The three relays on the left are what you would expect while the machine is running.

The left two relays (CR-1 and CR-2) are energized because the buttons are green.

Those relays are working correctly because there are 0 volts at terminals 14 so the contacts are closed.

This also tells us that Wire 121 is connected to common going all the way back through every machine to the master. If we were not seeing 0 volts on Wire 121 then likely one of the buttons are red on an upstream machine or one of the upstream machines has a bad relay at the CR-1 or CR-2 position.

If we were to jump Wire 121 to common it would signal to the rest of the system "all upstream machines have green buttons" regardless of the actual state of the buttons.

The third relay from the left (CR-ON) is energized to command this machine to run. The relay is correctly showing 24 volts on terminal 14 because the contact to terminal 11 is closed. This brings 24 volts to the run pin of the drives on this machine.

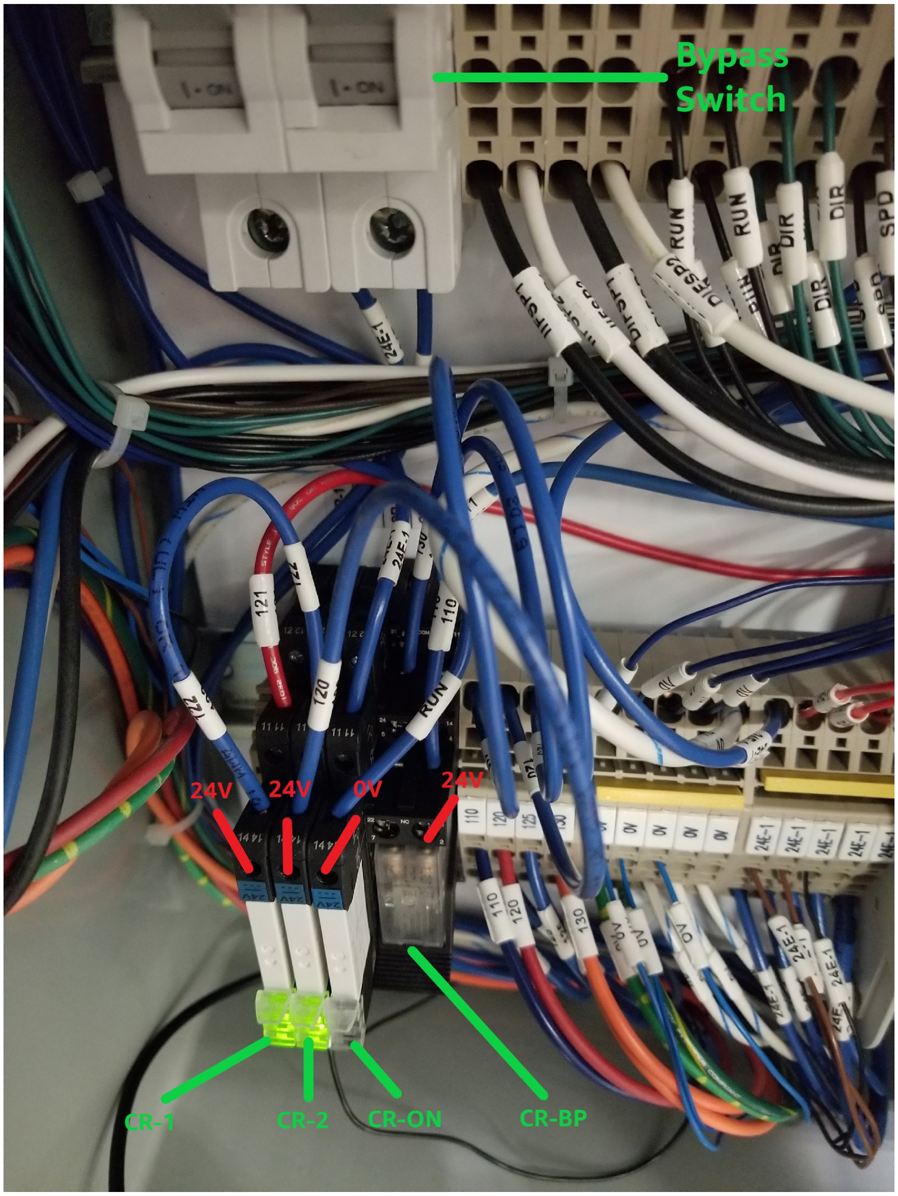
Notice the first relay on the right (CR-BP) is energized.

This only happens when the bypass switch (see the photo above) is flipped up and all the buttons on all the conveyor units are green.

We only do this on the very last machine and this is the very last machine.

Flipping the bypass switch up completes the circuit and sends the run signal on Wire 120 back down through all the machines on unobstructed line (orange Wire 130) to the master machine terminating at A2 of relay CR-SW which energizes CR-SW and commands all the machines to run by connecting Wire 110 to common which energizes all relays CR-ON in every machine throughout the system.

So signal starts at the Master with Wire 120 connected to common when CR-1 and CR-2 are energized as the buttons on the master machine change to green. Wire 120 connects to Wire 121 on the next machine through the yellow cable where the connection to Master common will continue on Wire 120 if this child machine also has green buttons. This continues along Wire 120 in every machine until we get to the last child (where if the bypass switch is flipped up and the button are green) then relay CR-BP is energized which connects common Wire 120 with orange Wire 130 going all the way back to terminal A2 of relay CR-SW in the master machine. When CR-SW in the master is energized the contact between Wire 110 and common is closed. And since Wire 110 is connected in parallel to terminals A2 of every CR-ON relay of every machine, all the machines start to run. So jumping Wire 110 to 0 volts causes all the machines to run regardless of what the buttons are doing on the machines. Only do this for testing – never in production.



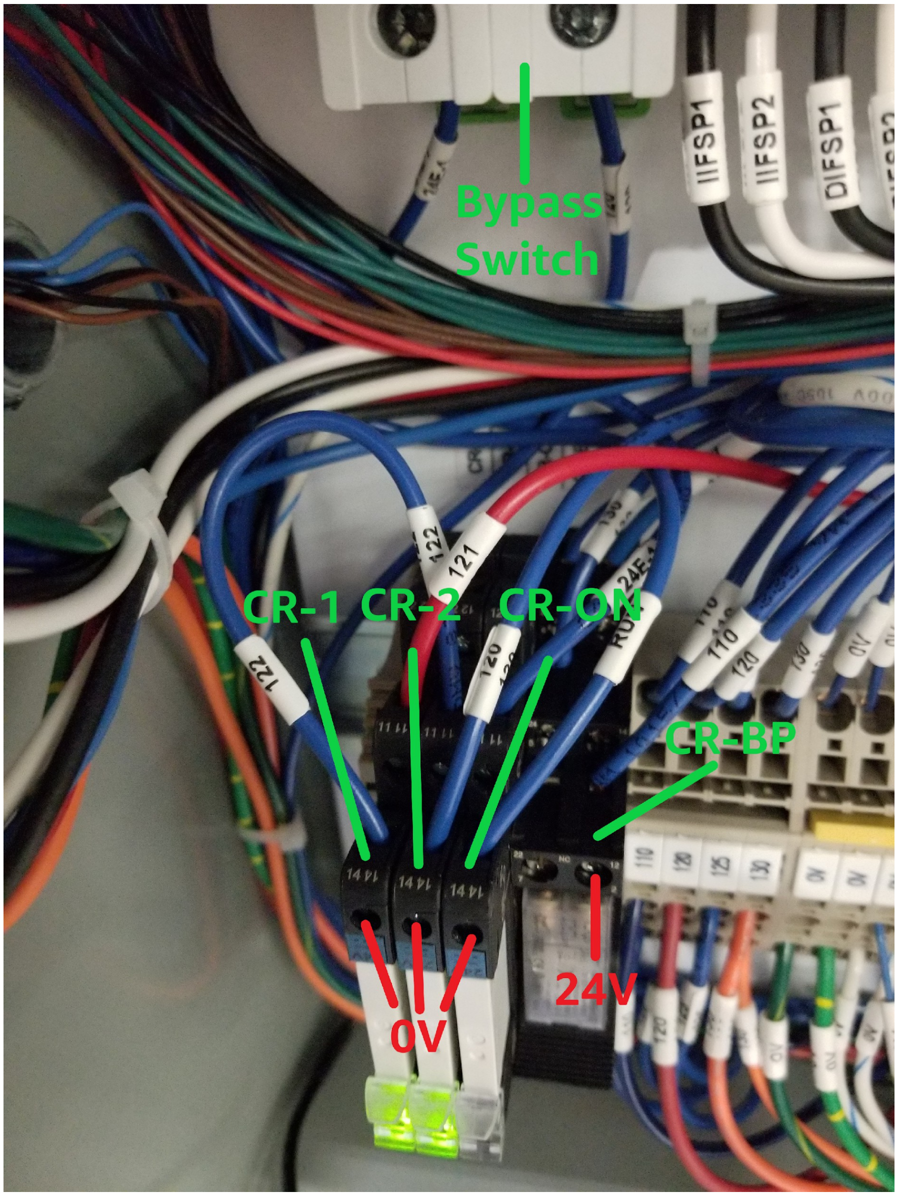
View of relays in the cabinet of the very last conveyor unit which is ready to go but the line is stopped.

Above we see what the last child in the line looks like when its own buttons are green but a previous machine has buttons red.

The first two relays from the left (CR-1 and CR-2) are energized because the buttons on this machine are green. But notice that terminals 14 on both relays measure 24 volts. That's because Wire 121 does not have continuity with common because the previous machine has a red button and which opens CR-1 or CR-2 on that machine to break continuity with common all the way back at the Master. And with Wire 120 connected to orange Wire 130 going all the way back to terminal A2 of relay CR-SW on the master, the wire is pulled to 24 volts through the coil which is not dropping any voltage due to lack of current. So if we are seeing 24 volts on those relays and if the relays are working, then we know that continuity to common is broken in an upstream machine on wire 120. If we are seeing 24 volts at these terminals then most likely we have a red button upstream. But if all the buttons are green then we have a broken wire, a bad relay upstream, or we have a bad (open) coil at CR-SW in the Master cabinet.

The third relay from the left (CR-ON) is not energized because none of the machines are permitted to run when there is a red button anywhere. Notice there are 0 volts at terminal 14 because that contact is open. So there is no 24 volt run signal going to the drives.

Since this is the last unit of the line, the first relay on the right (CR-BP) would be energized if all buttons on every machine including this one were green. But since a previous machine has a button red, this relay is not energized. So wire 130 is not connected to 120 which means that relay CR-SW in the master cabinet does not energize. And since CR-SW is what energizes all CR-ON relays in all machines by bringing Wire 110 to common, there are no 24 volt run signals going to any of the drives. Notice that Wire 110 is measuring 24 volts. That's because this wire is connected to terminal A2 of CR-ON in every machine. So if Wire 110 is not connected to common by CR-SW then it is being pulled to 24 volts by every CR-ON coil of every machine.



Relays in the cabinet of a midstream conveyor which is ready to go but the line is stopped downstream.

Above we see what a child unit looks like when it is in a ready to run condition (both buttons green) but a downstream unit has a button red.

Notice in the picture above, that there is 0 volts on terminals 14 of relays (CR-1 and CR-2) on the left when a downstream unit is stopped.

But notice in the previous photo that there is 24 volts on the same terminals when an upstream unit is stopped.

So this is one way we can zero in on a stopped unit even if its buttons are showing green. This can happen in the case where we have a bad power supply which is not supplying full voltage (this actually happened). It can also happen when relay CR-1 or CR-2 on any machine in the system is bad.

The thing to remember is that, assuming none of the machines are running, terminals 14 of the left two relays will carry 24 volts when an upstream machine is in a stopped condition and the same terminals will carry 0 volts when a downstream machine is in a stopped condition.

So if one of the machines is stopped without showing a red light then we can locate it by starting somewhere in the middle and checking voltage on those terminals. If we see 0 volts then we need to look downstream for the problem. If we see 24 volts then we need to look upstream for the problem.

Again we split the known problem area in the middle and repeat the procedure. Eventually we will come to the problem machine using this method.

So how does it all work?

There are 2 schematics we need to look at.

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Looking all the way at the bottom on the master drawing we see the relay (CR-SW)

This relay can be seen in the first photo of this document.

It is the first white relay on the right side of the relay pack.

When this relay is energized it's normally open contact (shown in the schematic on line 13 counting from the bottom) is closed. This connects Wire 110 to common which energizes every CR-ON relay in every machine because every one of those relays is connected to Wire 110 via the yellow Discharge Interface Cable which runs from one machine to the next. CR-ON in each machine brings 24 volts to the run pin of the MDR Drives when energized. This is seen in both schematics 13 lines from the top in about the center of the drawings.

So Jumping Wire 110 to ground on any machine should make all of them run regardless of the buttons on any of the machines. The buttons could be green or red it won't matter – all the machines will still run when Wire 110 of any machine is jumped to ground.

If there is a break in Wire 110 then the machines on the downstream side of the break will not run. This would be a good way to find out if any of the yellow cables are not connected and where exactly is the connection missing.

Relay CR-SW on the Master machine is energized when all the buttons on every machine are green.

Here is what makes that happen:

One side of CR-SW is connected to 24 volts.

The other side is connected to orange Wire 130 which goes directly to orange Wire 130 of the next machine via the yellow cable.

The next machine just extends the connection to orange Wire 130 going to the next machine in the same way.

So orange Wire 130 is just a very long wire running through each child machine but does not connect to anything inside those machines until it gets to a machine where the bypass circuit breaker/switch is flipped up.

Flipping up this bypass switch connects the common side (terminal A2) of relay coil CR-BP in this last child machine to Wire 120 via Wire 125 and also connects Wire 130 to Wire 120. And if all the push buttons on every machine are green then Wire 120 will be connected to common and this will energize CR-SW in the master machine which starts all the machines running as described above.

Wire 120 is connected to common in the master machine. Then it runs through the contacts of relays CR- 1 and CR-2 of every machine starting with the Master. These contacts only close to maintain continuity when the push buttons are green on that machine. Wire 120 extends to the next machine via the yellow cable and runs through the next set of relays CR-1 and CR-2 so on and so on until it gets to the machine where the bypass switch has been flipped. Here it is connected to orange Wire 130 which goes back to relay CR-SW in the master and connects it to common assuming all the push buttons are green. This energizes CR-SW which energizes the CR-ON in every machine to give the run signal to each drive.

If any one of the hundreds of relays CR-1 or CR-2 fail then terminal A2 of relay CR-SW will not have a connection to common and the machines will not run. Sadly, in this case there is no visual indication of which relay failed. The fastest way I can think of to find out would be to open a cabinet in the middle of the run and flip the bypass switch. If all the upstream machines run then we know that the problem is downstream. If they do not run then the problem is upstream. So split the problem area in the middle again and repeat the test continuing this way until the problem machine is found.

Troubleshooting Steps.

If one or a few adjacent rollers are not running then look for missing O-rings. If all the O-rings are in place then suspect the drive for those few rollers – see below for more on that.

First try turning the troublesome unit off and on again – That is usually all you have to do for most any problem. If that does not work then continue reading below.

If the drive light is green but the MDR for that drive will not run then check that the drive is getting a 24 volt run signal on pin 3. If the drive is getting the signal to run but the MDR is not running then you will need to replace the drive or replace the MDR. Plug in a new drive. If everything starts working then you will need to install a new drive. Otherwise you will need to change the MDR. If the problem is with the drive and the drive cannot be replaced immediately then unplug the cable going from the MDR to the drive. This will allow the MDR to turn freely and allow the other connected rollers to do the work of this one temporarily. If the MDR is not unplugged from the drive then the MDR will resist the other drives and cause the power supply to overheat which causes the entire line to shut down. Unplugging the drive from the cabinet is a good practice too but this will not allow the MDR to turn freely with the other rollers – you must unplug the MDR from the drive if you want the MDR to turn freely.

If a drive is showing a flashing red light then there is a minor fault. Likely turning the machine off and on again will solve the problem. It could be one of three things. 1. The MDR is trying to turn but is jammed and cannot move. 2. The MDR is overheating. 3 The drive is overheating. If reset mode is on (dip switch 4) then the drive will attempt to run 10 seconds after the temperature returns to the permissible operating range. Otherwise the machine will need to be turned off and on again to reset.

If machine runs noisy, a drive light sometimes goes dark, the line may stop and start uncommanded then likely what's happening is the power supply is overheating because one of the MDRs or one of the drives is bad. One of the MDRs will likely be hot but this one is not your problem. The MDR is hot because one of the other MDRs is not working and is resisting the work of the other MDRs. The noisy roller is likely the problem. The noisy one is the MDR that's not turning. Disconnect the MDR cable from the drive. This will allow the MDR to turn freely so that the other MDRs can take over the work of the noisy non-functioning MDR. If the noisy MDR is not unplugged from the drive then it will resist the other drives and cause the power supply to overheat which causes the entire line to shut down until the temperature drops to permissible levels in the power supply. Unplugging the drive from the cabinet is a good practice too but this will not allow the MDR to turn freely with the other rollers – you must unplug the MDR from the drive if you want the MDR to turn freely. Then when the associates are on break, plug in a new drive and see if it all works. If it does work then change the drive. If it still does not work then change the MDR.

If a drive is showing no light and it is receiving 24 volt power then the drive must be replaced. If the drive cannot be replaced immediately then unplug the cable going from the MDR to the drive. This will allow the MDR to turn freely and allow the other connected rollers to do the work of this one temporarily. If the MDR is not unplugged from the drive then the MDR will resist the other drives and cause the power supply to overheat which causes the entire line to shut down. Unplugging the drive from the cabinet is a good practice too but this will not allow the MDR to turn freely with the other rollers – you must unplug the MDR from the drive if you want the MDR to turn freely.

If the drive is showing a solid red light then check to be sure that the MDR is plugged into the drive. Aside from the connector at the drive, sometimes there is an extension cable between the drive and the MDR. If there is an extension cable then make sure that that is plugged in as well. If all the cables are connected between the drive and the MDR then it is possible that the power supply is not producing full voltage. If the power supply is producing the full 24 volts then either the MDR or the drive is bad. Change the drive first because that is the easiest thing to change. Don't pull the old drive out – just pull the cables out of the old drive and plug them into the new one and see if it works. If it does then change the drive, otherwise, change the MDR.

If a single drive is running in reverse then usually you can just power the machine off and on again to fix the problem. If this does not fix the problem then check that the directions pin on the drive is getting the same voltage signal that the other working drives are receiving. If the voltage is the same then change the drive. Otherwise look for a broken wire or loose connection.

If a portion of the line is running then most likely the bypass switch has been flipped up on the last running machine causing CR-BP to energize when all the upstream push buttons and the push buttons on this last running machine are green. A break in wire 110 or wire 111 which energizes the CR-ON relay in each machine could also be broken where the line stops running. Check for all of these conditions and make your repairs.

If the whole line will not run the most likely cause is a red button somewhere that is hard to see.

Look closely for the red button.

Also look closely for dark buttons. That is buttons that are neither red nor green. This is likely a machine that has been turned off or that has a bad power supply or perhaps no power at the AC receptacle. All the machines must be powered if the line is to run.

If all the buttons are green and yet nothing is running then the most likely cause is a loose or disconnected yellow cable that runs from machine to machine. If any yellow cables in the run are not connected to its downstream neighbor then none of the machines will run. Check that each and every yellow cable is connected to its downstream neighbor.

If all the yellow cables are connected then check the very last machine to see that the Bypass switch is flipped to the up position. Some of the lines split and a package can go in one of two directions. In this case there will be two machines that are at the end of the line and each of these machines should have the Bypass switch flipped up.

If the Bypass switch is flipped up on the very last machine then jump Wire 110 to common (0v). This should cause all the machines to run because bringing Wire 110 to common bypasses the start/stop circuit and commands all the machines to run regardless of the state of the buttons. Wire 110 could be jumped to common on any machine because all the machines are connected together on this wire via the yellow cable but it’s better to jump Wire 110 to common at the Master unit just in case the wire is broken somewhere. Jumping Wire 110 to common bypasses the stop/start circuit. Only do this when troubleshooting – never in production. This should cause CR-ON in every machine to energize sending a run signal to every drive. So all machines should be running after jumping Wire 110 to common.

With Wire 110 jumped to common, walk along the line to be sure that all the machines are running.

If any of the machines are not running then you now know which machine is causing the problem. Look in the cabinet of that machine. The problem should be obvious. Check for loose wires, bad relays, bad buttons, or a bad power supply. There isn’t much else that can go wrong with these machines.

If all the machines run when Wire 110 is jumped to common then the problem is with the start/stop circuit of one of the machines. The problem could be a bad CR-1 or CR-2 relay in any of the machines or a loose connection on Wire 120, Wire 121, or Wire 130 which could happen in any of the machines as well. It could also be a bad relay CR-BP on the very last machine which is failing to connect Wires 120 and 130 together via Wire 125.

It would seem to be difficult to find which machine is causing the problem but actually it’s not too difficult. The following is the procedure.

First open the Master cabinet and flip the Bypass switch up. Now run the Master unit. The master unit should run if its buttons are green while all the other machines remain off. If the Master does not run then the problem is with the master unit so check for the problems listed above.

If the Master runs then flip the Bypass switch down again and then go to a machine in the middle of the run and flip its bypass switch up. This machine is now the last machine in the run. All the machines downstream have been removed from the system when the bypass switch was flipped up on this machine. So none of the downstream machines will run when the system is started but if the problem with the start/stop circuit is on the downstream side, those problems will no longer prevent the upstream side from running. So if the upstream side now runs then we know the problem is downstream. If the upstream side still does not run then we now know that the problem with the start/stop circuit is with the upstream side.

Flip the Bypass switch down again on this machine. Then go to the middle of the new problem area and flip the Bypass switch up and repeat the test. Again, if the upstream side runs then the problem is downstream. If it does not run then the problem is still upstream. Flip the Bypass switch down again and repeat this process until the troublesome unit is found.

Once the problem unit is found, look for a bad relay at CR-1 or CR-2. If these are good, look for loose wires 120, 125, or 130. If continuously dividing the start/stop circuit has brought you to the very last machine in the run then be sure to check that relay CR-BP is functioning.

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